

COMPETITOR GUIDE - v1.0.0

RANK :: CHALLENGE

Node.js Coding Challenges

10 CHALLENGES - 2,650 POINTS - NODE.JS ONLY

A progressive, code-fixing Node.js challenge that takes you from modules and the file system all the way to async, streams, and Express. Each level unlocks the next. Fix the code, earn the points, climb the developer ranking.

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What is Rank Challenge?

Rank Challenge is a code-fixing competition for Node.js developers. Every level hands you a small, broken Node.js script. Your job is to fix it so it behaves correctly - and the platform grades your code automatically.

There are 10 challenges worth a combined 2,650 points. They form a single progression: each level you solve unlocks the next, and your total score places you on the developer ranking.

Every challenge is Node.js

The whole track is server-side JavaScript. You work in .js files inside an in-browser editor. Topics climb from the fundamentals to real backend patterns:

- Modules - exporting and importing code
- File system - fs/promises with async/await
- HTTP - Express routes and res.json
- Async - callbacks to async/await, try/catch error handling
- Advanced - process.env config, streams, and EventEmitter

You only need to know Node.js

No front-end or database knowledge is required. If you can write a function and use async/await, you have everything you need.

How grading works

When you press Submit, the server parses your code against a set of checks for that level (for example: "uses await", "calls res.json", "has try/catch"). All checks must pass to clear the level.

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Install Node.js

Rank Challenge runs on Node.js (which includes npm). You need Node 18 or newer; the latest LTS release is recommended.

Step 1 - Check if you already have it

Open a terminal (Terminal on macOS/Linux, or PowerShell / Command Prompt on Windows) and run:

```
node -v  
npm -v
```

If both print a version (and node -v is v18 or higher), you're ready. Otherwise install or update Node below.

Step 2 - Download & install

Official installer (pick LTS): <https://nodejs.org/en/download>

Run the installer, accept the defaults, then reopen your terminal and rerun node -v to confirm.

```
# macOS (Homebrew)  
brew install node  
# Windows (winget)  
winget install OpenJS.NodeJS.LTS  
# Debian / Ubuntu  
sudo apt-get install -y nodejs npm
```

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Run the Challenge

With Node installed, you start the platform from the project folder in three commands.

Step 1 - Open the project

```
cd path/to/node
```

Step 2 - Install dependencies (first time only)

```
npm install
```

This installs the frontend and backend packages (npm workspaces). It only needs to be done once.

Step 3 - Start it

```
npm run dev
```

This launches the frontend (<http://localhost:5173>) and the backend (<http://localhost:3001>) together.

Step 4 - Open & log in

Open <http://localhost:5173>, click Continue, then enter your 6-digit access code. A successful login starts a 1-hour session.

Login codes are issued by the 2FA service your organiser runs (<http://localhost:3000>). Each code is single-use and expires quickly.

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The 10 Challenges

All ten levels are Node.js. Solve them in order - each unlocks the next.

#	TITLE	WHAT YOU'LL FIX	POINTS
01	Export a Function	module.exports	100
02	Read a File	fs/promises with await	150
03	JSON Route	Reply with res.json	200
04	Parse the Body	Read req.body	250
05	Go Async	Callback to async/await	250
06	Handle Errors	try/catch around await	300
07	Read Config	process.env with a fallback	300
08	Stream a File	createReadStream + .pipe	350
09	Emit Events	EventEmitter on / emit	350
10	Express Endpoint Finale	async + try/catch + res.json	400

Each level tells you exactly what to do

Every challenge has a description, a list of objectives, and optional hints. The objectives map directly to the checks the grader runs.

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Scoring & Rules

Points

Each level is worth the points shown above, from 100 (Level 01) up to 400 (the finale) - 2,650 in total. Points are banked the moment all of a level's checks pass.

Hints cost points

Revealing a hint subtracts a fixed penalty from that level's score (around 15 points). Use them only when stuck.

Unlock chain

Levels unlock in sequence: clear Level 01 to open Level 02, and so on to the finale. Your progress is saved within your session.

Session window

A login is valid for 1 hour. When the timer in the top bar runs out you return to the login screen and need a new code.

Submission limit

Submissions are rate-limited to 20 per minute to keep things fair - no penalty, you just wait a moment if you hit it.

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The Interface & Finishing

Top bar

- Server status - confirms the backend is reachable
- Session timer - time left in your 1-hour window
- Score & progress - your points and solved/total count
- Challenge Map - jump back to the level grid
- Finish - end your attempt

Editor & results

The left panel is the code editor (your .js file). The right panel is a results console - press Submit to grade your code and see which checks passed.

Use Submit to grade your answer, Hint to reveal a hint (costs points), and Reset to restore the original starter code (.js file).

Finishing the challenge

When you're done, press Finish in the top bar to see a summary of how many levels you solved and your final score, then confirm to end the session.

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Tips & Quick Reference

Strategy

- Read the objectives first - they mirror exactly what the grader checks.
- Match the exact API the objective names - res.json, fs/promises, EventEmitter, and so on.
- Save hints for later - try the fix yourself first; hints cost points.
- Submit early, submit often - failed checks tell you what's still wrong.
- Reset if you get lost - it brings back the original starter.

Quick command reference

```
node -v          # check Node version (need 18+)
npm install      # one-time dependency install
npm run dev      # start frontend + backend
Ctrl + C        # stop both servers
```

Good luck - fix the code, earn the points, climb the ranking.